Lecture 13: Lagrange Multipliers

Min/Max a function $f(x,y,z)$ where x,y,z aren't independent $g(x,y,z) = c$

Example: point closest to origin on hyperbola $xy=3$

Minimize: $f(x,y) = x^2+y^2$ Subject to constraint $\underbrace{xy}_{g(x,y)}=3$

Observe: at the minimum, level curve of f is tangent to hyperbola $g=3$

How to find (x,y) where level curves of f and g are tangent to each other! When this happens, $\nabla f \parallel \nabla g$

So $\nabla f = \lambda \nabla g$

Min/Max System of equations $\nabla f = \lambda \nabla g$ $\begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$ constraint $g=c$

$f=x^2+y^2$, $g=xy$ $\begin{cases} 2x=\lambda y \\ 2y=\lambda x \\ xy=3 \end{cases} \iff \begin{cases} 2x-\lambda y=0 \\ \lambda x-2y=0 \end{cases} \iff \underbrace{\begin{bmatrix} 2 & -\lambda \\ \lambda & -2 \end{bmatrix}}_{M} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Trivial solution $(0,0)$ does not solve $xy=3$

Other solutions exist only if $\det(M)=0$

$\begin{vmatrix} 2 & -\lambda \\ \lambda & -2 \end{vmatrix} = -4+\lambda^2 = 0 \iff \lambda^2=4 \iff \lambda = \pm 2$

Two cases

$\lambda=2$ $x=y$ $x^2=3$ $(x,y)=(\sqrt{3},\sqrt{3})$ or $(\sqrt{-3},\sqrt{-3})$	$\lambda=-2$ $x=-y$ $-x^2=3$ No solutions

Why is this method valid?

At constrained min/max, in any direction along level $g=c$, the rate of change of f must be 0.

For any $\hat{u}$ tangent to $g=c$, we must have $\frac{df}{ds}=0$

$\nabla f \cdot \hat{u} = \frac{df}{ds}$

So any such $\hat{u}$ is $\perp \nabla f$

So $\nabla f \perp$ level set of g, we know that $\nabla g \perp$ level set of g, so $\nabla f \parallel \nabla g$

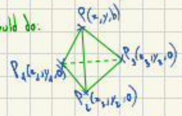
To find min (or max)

We compare values of f at the various solutions to Lagrange Equation.

Advanced Example: We want to build a pyramid with given triangular base & given volume.

Minimize total surface area. Vol: $\frac{1}{3}$ Area(base) $\cdot \underbrace{\text{height}}_{\text{fixed}=h}$

Could do:



If we express the area in terms of (x,y), we can't solve minimum

From above:

Let u_1, u_2, u_3 = dist. from Q to sides

height of faces: $\sqrt{u_1^2+h^2}, \sqrt{u_2^2+h^2}, \sqrt{u_3^2+h^2}$

Side area $= \frac{1}{2}a_1\sqrt{u_1^2+h^2} + \frac{1}{2}a_2\sqrt{u_2^2+h^2} + \frac{1}{2}a_3\sqrt{u_3^2+h^2} = f(u_1,u_2,u_3)$

Cut base into 3 pieces $\Rightarrow$ Area(base) $= \frac{1}{2}a_1u_1 + \frac{1}{2}a_2u_2 + \frac{1}{2}a_3u_3$

$\Delta f = \lambda \nabla g$: $\overbrace{\frac{\partial f}{\partial u_1} = \cancel{\frac{1}{2}a_1}\frac{u_1}{\sqrt{u_1^2+h^2}}}^{\nabla f} = \lambda \cdot \overbrace{\cancel{\frac{1}{2}a_1}}^{\nabla g}$ So $u_1=u_2=u_3$ Q should be the incenter (equidistant from all sides)

$\frac{\partial f}{\partial u_2} = \cancel{\frac{1}{2}a_2}\frac{u_2}{\sqrt{u_2^2+h^2}} = \lambda \cdot \cancel{\frac{1}{2}a_2}$

$\frac{\partial f}{\partial u_3} = \cancel{\frac{1}{2}a_3}\frac{u_3}{\sqrt{u_3^2+h^2}} = \lambda \cdot \cancel{\frac{1}{2}a_3}$